CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 12

ECHO AND CHAT CLIENT-SERVER APPLICATIONS USING TCP/UDP SOCKETS

Aim

To develop and test echo and chat applications between a client and server using TCP/UDP sockets.

PART A – ECHO CLIENT-SERVER

Algorithm

1. Create a server socket and associate it with the required address and port.
2. For TCP communication, listen for incoming connections and accept a client connection.
3. The client sends a message to the server.
4. The server receives the message and sends the identical message back to the client.
5. The client receives and displays the returned message.

Input / Output

Input: A text message typed by the client.
Output: The server returns the same message, which is then shown by the client.

PART B – CHAT CLIENT-SERVER

Algorithm

1. Create and set up the sockets for both server and client.
2. Establish a connection when using TCP, or use **sendto()** and **recvfrom()** for UDP communication.
3. Let the client and server send and receive messages in turn.
4. Display each message received at the respective endpoint.
5. End the chat when either participant enters **exit**.

Input / Output

Input: Messages typed by the client and server.
Output: Both sides display the messages exchanged until the session is stopped.

Procedure

1. Save the client and server code in separate Python files, or create one program that allows the user to select a role.
2. Start the server program first.
3. Run the client in a separate terminal, either on the same computer or on a reachable host.
4. Enter a message and check the echo or chat response.
5. Type **exit** to close the chat session.

Observation

The echo server sent back the same message received from the client. In the chat application, messages were exchanged between the client and server through socket communication.

Result

Thus, echo and chat client-server applications were implemented and tested using socket programming.